Data Science

BY THE END:

You can explain what data science is and what data scientists do.

Overview

A mobile company notices that thousands of customers have stopped buying their monthly bundles. The records show what happened, but the company needs to understand the pattern, test possible reasons, and decide what evidence would support a solution. This kind of problem without one simple answer is where **data science** becomes useful.

What is data science?

Data science combines statistics, programming, domain knowledge, and data analysis to find useful answers and build solutions using data.

Data science may use statistical analysis, experiments, simulations, careful exploration, or machine learning.

Each part contributes something different:

PART	WHAT IT CONTRIBUTES
Statistics	Ways to measure patterns, uncertainty, and evidence
Programming	A way to work reliably with large or complex data
Domain knowledge	Understanding the real-world problem and its limits
Data analysis	Ways to explore data and explain what it shows

Knowing the business matters as much as knowing the numbers. A pattern in mobile data means little if the person studying it does not understand bundles, prices, customers, or how the records were made.

What data scientists do

Data scientists often work on problems such as:

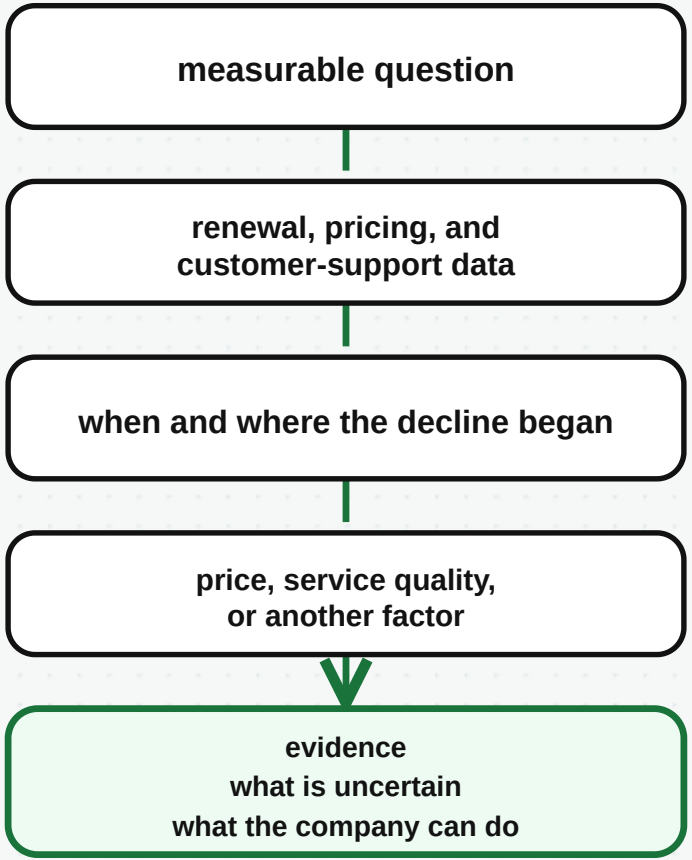
- Predicting what people will need
- Detecting fraud
- Testing ideas and checking the results
- Studying complicated customer behavior
- Building data products or predictive models

The work usually begins with a question, not a model. A data scientist decides what evidence is needed, checks whether the right data exists, chooses the right method, tests the result, and explains its limits.

Example

For the falling renewals, a data scientist might:

1. Turn “customers are leaving” into a measurable question
2. Combine renewal, pricing, and customer-support data
3. Look for when and where the decline began
4. Test whether price, service quality, or another factor best explains it
5. Show the evidence, what is uncertain, and what the company can do



If the company later asks which individual customers are likely to leave next month, the problem has changed. It now needs a prediction. That is where machine learning may enter the project.